

# Alok Raj

+91-6206369741 | rajalok10375@gmail.com | LinkedIn | GitHub | LeetCode

## EDUCATION

### VIT Bhopal University

B.Tech in Computer Science and Engineering (Cloud Computing and Automation)

Sep 2023 – May 2027

CGPA: 8.93/10

## TECHNICAL SKILLS

**Languages:** Python, Java, JavaScript, SQL

**Systems and Backend:** Distributed Systems, Microservices, REST APIs, FastAPI

**Cloud and Infrastructure:** AWS (Lambda, API Gateway, DynamoDB, S3, EventBridge, Step Functions, CloudWatch, IAM, Rekognition, SES)

**AI and GenAI:** RAG Pipelines, LLM Integration, Prompt Engineering, Amazon Bedrock, Pinecone, OpenSearch, Vector Databases

**Databases:** DynamoDB, MongoDB, MySQL

**Frontend:** React.js, HTML5, CSS3

**DevOps and Tools:** Git, GitHub Actions, CI/CD, Docker, Postman, n8n

**Core CS:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks

## EXPERIENCE

### Cloud Engineering Intern – Backend Systems

Aug 2025 – Jan 2026

IPAGE Group

Remote

- Designed and deployed 30+ serverless microservices on AWS Lambda to automate an 8-step company registration and AI content generation workflow for dronetv.in, supporting 5,000 concurrent users and 500–1,000 daily registrations.
- Reduced AI content generation costs by 88% by implementing intelligent model routing that dynamically selected GPT-3.5 or GPT-4 based on request complexity, eliminating unnecessary high-cost inference.
- Built a three-stage approval pipeline using DynamoDB state tracking (draft, under review, approved) with automated Nodemailer notifications triggered at each transition, removing all manual follow-up.
- Developed an admin backend with JWT authentication and role-based access control, enabling structured approval and publishing workflows with all actions logged for audit traceability.

## PROJECTS

### MarketMind | React, FastAPI, Pinecone, LangChain, FinBERT, Groq LLaMA | GitHub

Feb 2026 – Mar 2026

- Collaborated in a 3-member team to build a RAG-powered stock intelligence platform ingesting data from five live financial sources using semantic retrieval and real-time market signal analysis.
- Designed and built the Finfluencer Fact-Checker feature – a pipeline that extracts financial claims from YouTube transcripts chunk-by-chunk, verifies each against live market data, and returns structured verdicts (Verified, Misleading, False, Unverifiable) with a video-level risk score.
- Developed the React frontend and integrated backend APIs, enabling seamless end-to-end financial claim verification within the platform.

### Sahayak AI | AWS Lambda, Bedrock, EventBridge, S3, DynamoDB, OpenSearch, Python, React | GitHub

Sep 2025 – Jan 2026

- Built an AI-powered classroom automation platform for Indian middle schools (Grades 6–8) using 20+ AWS Lambda functions and Amazon Bedrock (Nova, Titan, Deepseek), automating lesson scheduling and doubt resolution end-to-end.
- Engineered an NCERT data lake on S3 with OpenSearch Serverless vector indexing and a RAG pipeline, reducing query response latency by **70%** through semantic retrieval over structured educational content.
- Implemented an EventBridge-driven daily pipeline that enriches teacher-uploaded lessons with Bedrock-generated questions and YouTube recommendations, delivers content to students automatically, and dispatches automated student notifications via Nodemailer.

### SafeX 3.0 | AWS Rekognition, Lambda, API Gateway, S3, MongoDB, Python, React | GitHub

Feb 2025 – Apr 2025

- Designed a 3-factor authentication system combining bcrypt password verification, AWS Rekognition facial recognition at **99% accuracy**, and OTP, orchestrated through 8 REST API endpoints on API Gateway with face data in S3 and user metadata in MongoDB.
- Implemented IAM role-based access control, signed S3 URLs for secure file vault access, and CloudWatch logging across all Lambda functions to maintain a complete audit trail.

## CERTIFICATIONS AND ACHIEVEMENTS

- AWS Certified Solutions Architect – Associate** – Jun 2026 (Score: 893/1000)
- AWS Certified Cloud Practitioner** – Apr 2026 (Score: 1000/1000)
- IBM DevOps and Software Engineering Professional Certificate** – Feb 2026
- Top Performing Candidate** – ET AI Hackathon 2026, The Economic Times (May 2026)
- Winner** – Nebula Nexus Hackathon, MIT Jaipur (Jul 2025)